AgriSoko Pocket G11 - Project Report

Group code: G11 | Version: 1.0.0 | Repository: cxnisme/agrisoko-pocket-g11

Project overview

AgriSoko Pocket G11 is an Expo/React Native Android app paired with a FastAPI service. It supports member task lookup, explainable profile matching, and a persistent offline fallback. The app presents GitHub usernames and assigned task areas until each member approves any personal profile claims.

System design

The mobile client uses typed bottom-tab navigation, query validation, request timeouts, explicit source badges, and persistent AsyncStorage caching for directory, profiles, and successful queries. When the API cannot be reached, it uses bundled GitHub task records and local token matching. The API exposes health, model-info, member-directory, profile, and ask endpoints. Ask uses deterministic TF-IDF/cosine-style term weighting with exact name and alias boosts and a confidence/margin guard.

Validation performed

Backend: 13 tests passed in the final run, including request bounds, CORS preflight, member lookup, offline task matching, unknown profiles, empty directory, and unrelated queries. Mobile: 7 Jest tests passed for bundled fallback, directory/profile/query caching, offline matching, no-match behavior, malformed responses, and server errors. TypeScript check passed; Expo dependency check passed; Expo Doctor passed 18/18; Android JavaScript bundle export succeeded.

Release and evidence limitations

A cloud EAS Android APK build has been queued. The final signed APK must be copied to build/AgriSokoPocket_G11.apk only after EAS reports success and the downloaded APK is verified. No physical Android device test, actual demo recording, external feedback, or member profile sign-off is claimed by this report. Confirm those items before submission. The public API deployment URL is also not configured; use EXPO_PUBLIC_API_URL for a deployed HTTPS service, otherwise the APK retains its offline fallback.

Assignment layout

The requested COMBO_A3_G11 folder uses README.md, mobile/, ai-api/, build/, docs/COMBO_A3_G11_REPORT.pdf, and evidence/ with disclosure, contribution, test, and feedback records. The demo video is intentionally not synthesized; capture it on a real device.